3 A rollercoaster carriage of mass 280 kg travels over a raised section of track, as illustrated in Figure 3.1.

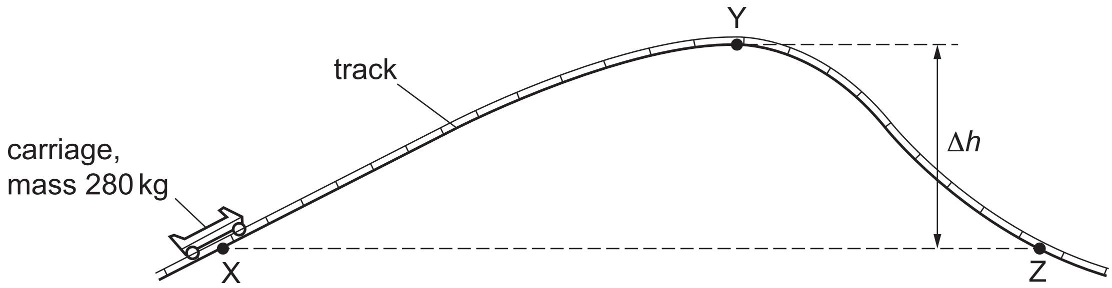

Figure 3.1

At position X, the carriage has kinetic energy 16.1 kJ. At position Y, the carriage has kinetic energy 1.3 kJ. The length of track between X and Y is 23 m.

In moving from X to Y the carriage does 5.1 kJ of work against resistive forces.

(a) **(i)** Calculate the average resistive force acting on the carriage between X and Y.

average force = ... N [2]

(ii) Calculate the change in height Δh between X and Y.

Δh = ... m [3]

(b) The carriage travels further along the track to point Z.

State and explain whether the speed of the carriage at Z is greater than, less than or the same as the speed of the carriage at X.

....

....

.... [1]

[Total: 6]